

PRACTICAL 1

Goal: To familiarise yourself with OrCAD Schematic Capture and Pspice, which will be used to simulate DC circuits.

Tasks:

Practical 1 consists of three parts:

1. A pre-practical assignment that must be completed individually in the week that precedes practical 1.
2. A pre-practical test that must be completed at the start of the practical session.
3. The practical itself that will be completed in groups of two during the practical session.

TASK 1

In order to complete the assignment, you will need to use the OrCAD software. A tutorial has been created that you can use to both install the software and familiarise yourself with the program's functionality.

Go to the **Practical 1 Tutorial** link on EBN111's clickUP page to download the document.

Be sure to complete this tutorial before starting with the assignment. Should you already be familiar with OrCAD then you may skip the tutorial.

The pre-practical assignment can be found on EBN111's clickUP page.

You are required to download this assignment.

Start the pre-practical assignment when you are ready, complete it and bring your completed assignment to the practical session.

TASK 2

The pre-practical test will be written at the start of your practical session and the questions will test you in the following areas:

- a) Information provided during the pre-practical lecture
- b) Information contained in the slides provided for practical 1
- c) Information obtained from completing the tutorial
- d) Information obtained from completing the pre-practical assignment itself.

The test will be answered individually and will take up 30 minutes of the practical session. After completing the test, you will move on to the practical itself.

Note that this test is NOT 'open-book'. Hence you will be required to complete it without reference to any of the study material.

TASK 3

The practical itself must be completed during the practical session. Students may work in groups of 2.

Print out the Practical 1 Guidelines (found on clickUP) and bring your printout to the practical session. These are required for you to complete the practical.

Follow the instructions, perform the simulations and then fill in your answers.

After you have completed the practical and have filled in all the values, go onto clickUP and complete the Practical 1 Evaluation.

You will not submit your hardcopy Practical 1 Guidelines, but the answers which you have noted here must be submitted to the Practical 1 Evaluation on clickUp. This evaluation will only be open during the time of your practical session, therefore must be completed within these three hours.

CONCLUSION

In order to pass the practical, you have to complete and bring the pre-practical assignment to the practical, and also submit the following clickUP evaluations:

- 1) Practical test.
- 2) Practical 1 evaluation.

Marks will be released at the end of the week, once all groups have completed their practical sessions.